

Hardware Info Cheat Sheet

Find out what a machine is made of: CPU, memory, cards, disks, network ports, serial numbers and OS.

Using it

To see	Run	To see	Run
CPU	lscpu , nproc (CPU count)	Disks	lsblk , lsblk -d -o NAME,MODEL
Memory	free -h , lsmem (online total)	Network ports	ip -br link , ethtool eno1
PCI cards	lspci , lspci -k , lspci -nn	Temperatures	sensors (lm-sensors package)
USB devices	lsusb (-t: as a tree)	OS and kernel	hostnamectl , uname -r
Serials, RAM	sudo dmidecode -t system/memory	Everything	inxi -Fz (inxi package)

Reading lscpu (trimmed)

```
Architecture:      x86_64 ← aarch64 = ARM
CPU(s):            16 ← logical CPUs; = nproc
  Model name:      Intel(R) Xeon(R) Silver 4208
    Thread(s) per core: 2 ← 2 = hyper-threading on
    Core(s) per socket: 8 ← real cores per chip
    Socket(s):      1 ← CPU chips: 1 x 8 x 2 = 16
  Virtualization:  VT-x ← can host VMs (AMD: AMD-V)
```

Reading hostnamectl

```
Static hostname: web1
  Virtualization: kvm ← only shown inside a VM
Operating System: Ubuntu 24.04.1 LTS ← os-release
  Kernel: Linux 6.8.0-45-generic ← uname -r
Hardware Vendor: QEMU ← real box: Dell Inc.
Hardware Model: Standard PC_Q35 + ICH9, 2009_
real box: PowerEdge R640, no Virtualization line
```

Cards and disks

```
$ lspci -k ← -nn adds [class] and [vendor:device] IDs, e.g. [8086:1521]
03:00.0 Ethernet controller: Intel Corporation I350 Gigabit Network Connection (rev 01)
  Kernel driver in use: igb ← the driver running it

$ lsblk -d -o NAME,SIZE,MODEL,SERIAL,TRAN ← -d: whole disks, no partitions
NAME        SIZE MODEL                                SERIAL                                TRAN
sda          894.3G MZ7KH960HAJR0D3                S47NNA0R123456                sata ← TRAN: sata, sas, nvme, usb
nvme0n1      476.9G SAMSUNG MZVL2512HCJQ-00B00    S675NX0R654321                nvme
```

dmidecode: serials, DIMM slots, BIOS (needs sudo)

Command	Look for	Command	Look for
-t system	Maker, Product Name, Serial Number	-t bios	Vendor, Version, Release Date
-t memory	Per slot: Locator, Size, Type, Speed	-s system-serial-number	Just the serial (service tag)
	No Module Installed = empty slot	-t memory grep -c DDR	Rough count of fitted DIMMs

Network ports

```
$ ip -br link
eno1          UP                d0:94:66:1a:2b:3c <BROADCAST,MULTICAST,UP,LOWER_UP>
eno2          DOWN            d0:94:66:1a:2b:3d <NO-CARRIER,BROADCAST,MULTICAST,UP> ← no cable

$ sudo ethtool eno1
  Speed: 1000Mb/s ← 1 Gb/s; 10000Mb/s = 10 Gb/s. Lower than expected = bad cable
  Duplex: Full ← Half = negotiation problem
  Link detected: yes ← cable plugged in, switch port alive

$ ethtool -i eno1
driver: igb ← kernel driver
firmware-version: 1.63, 0x800009fa ← NIC firmware (compare to vendor's latest)
```

Handy recipes

Command	Does
sudo inxi -Fz	Full summary, serials and MACs hidden: safe to paste in a ticket
sudo sensors-detect , then sensors	Set up and read CPU/board temperatures and fan speeds
lspci -k grep -A3 -i net	Network cards and the driver each one uses

Remember: most of these need **sudo** for full detail (serials, DIMMs, firmware). **inxi -Fz** gives a shareable summary with serials hidden.